BEE 4750 Homework 4: Linear Programming and Capacity Expansion

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Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `c:\Users\adric\OneDrive - Cornell University\Documents\3. Env Sys A
```

```julia
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
using Statistics
```

Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$\begin{aligned}
\max_{x,y} \quad & 3x + 2y \\
\text{subject to:} \quad & x + 4y \leq 16 \\
& x - 2y \leq -1 \\
& 0 \leq x \leq 4 \\
& 1 \leq y \leq 4.
\end{aligned}$$

Split constraints into:

$$\begin{aligned}
\max_{x,y} \quad & 3x + 2y \\
\text{subject to:} \quad & x + 4y \leq 16 \\
& x - 2y \leq -1 \\
& 0 \leq x \\
& x \leq 4 \\
& 1 \leq y \\
& y \leq 4.
\end{aligned}$$

Lagrangian function:

$ L = 3x - 2y + *{1} (x + 4y - 16)* - {2} (x - 2y + 1) - *{3} (-x)* - {4} (x - 4) - *{5} (1-y)* - {6} (y - 4) $

$ dL/dx = 3 - *{1}* - {2} + *{3}* - {4} = 0 $
$ dL/dy = 2 - 4 *{1} + 2* {2} + *{5}* - {6} = 0 $

Constraints 1 and 2:

$ x + 4y = 16 $
$ x - 2y = -1 $

2 into 1:
$ x = 2y - 1 $
$ 2y -1 + 4y = 16 →6y = 17 →y = 17/6 = 2.833 →x = 16 - 4*17/6 = 4.667$

Checking remaining constraints:
(3) $ x > 0 $
(4) $ x < 4 $ failed

Constraints 1 and 4

$ x + 4y = 16 $
$ x = 4 $

3 into 1:
$ 4 + 4y = 16 →y = 3 $

Checking with remaining constraints:
(2) $ 4 - 2*3 = 4 - 6 = -2 < -1 $
(3) $ x > 0 $
(5) $ y > 1 $
(6) $ y < 4 $

Plugging into lagrangian derivatives:
$ 3 - *{1} +* {3} = 0 2 - 4 _{1} = 0 $

2 into 1: $ *{1} = 1/2 3 - 1/2 +* {3} = 0 → _{3} = 5/2 $

Both lambdas > 0, therefore viable case.
Function:
$ f(x, y) = 3*4 + 2*3 = 12 + 6 = 18 $

Constraints 2 and 4:

$ x - 2y = -1 $
$ x = 4 $

3 into 2:
$ 4 - 2y = -1 →y = 5/2 $

Checking with remaining constraints:
(1) $ 4 + 4*5/2 = 4 + 10 = 14 < 16 $
(4) $ x = 4 > 0 $

(5) $ y = 5/2 > 1 $
(6) $ y = 5/2 < 4 $

Plugging into lagrangian derivatives:
$ 3 - {2} + {3} = 0   2 + 2 _{2} = 0 $

2 into 1:
$ {2} = 1/2 $
$ 3 - 1/2 + {3} = 0 → _{3} = 5/2 $
Both lambdas > 0, therefore viable case.
Function:
$ f(x, y) = 34 + 25/2 = 12 + 5 = 17 $

Constraints 1 and 6:

$ x + 4y = 16 $
$ y = 4 $

6 into 1:
$ x + 4*4 = 16 →x + 16 = 16 →x = 0 $

Checking with remaining constraints:
(2) $ 0 - 2*4 = -8 < 16 $
(3) $ x = 0 $
(4) $ x = 0 < 4 $
(5) $ y = 4 > 1 $

Plugging into lagrangian derivatives:
$ 3 - {1} = 0   2 - 4 {1} - _{6} = 0 $

1 into 2: $ _{1} = 3 $ $ 2 - 4*3 - {6} = 0 → {6} = 2 - 12 = -10 $

Lambda 6 is negative, failed.

Constraints 2 and 6:

$ x - 2y = -1 $
$ y = 4 $

6 in to 1:
$ x - 2*4 = -1 →x = 7 $

Checking with other constrains:
(4) $ x = 7 > 4 $ failed

Constraints 5 and 6:

$ x = 4   y = 4 $

Checking with remaining constrains:
(1) $ 4 + 4*4 = 4 + 16 = 20 > 16 $ failed

Summary:
(4, 3), f(4, 3) = 18
(4, 5/2) ,f(4, 5/2) = 17

Optimal solution:
x = 4, y = 3

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs $70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost ($/ha-yr)	Selling Price ($/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear**.

Soybeans

Rate = 0 kg/ha

yield = 2900 kg/ha
Revenue $ = 2900*0.36 = 1026 $*
*Cost $ = 350 + 70*0 = 420 $
Profit/ha $ = 1044 - 350 = 694 $

Rate = 1 kg/ha

yield = 3800 kg/ha
Revenue $ = 3800*0.36 = 1368 $*
*Cost $ = 350 + 70*1 = 420 $
Profit/ha $ = 1368 - 420 = 948 $

Rate = 2 kg/ha

yield = 4400 kg/ha
Revenue $ = 4400*0.36 = 1584 $*
*Cost $ = 350 + 70*2 = 490 $
Profit/ha $ = 1584 - 490 = 1094 $

Wheat

Rate = 0 kg/ha

yield = 3500
Revenue $ = 3500*0.27 = 945 $*
*Cost $ = 280 + 70*0 = 280 $
Profit/ha $ = 945 - 280 = 665 $

Rate = 1 kg/ha

yield = 4100
Revenue $ = 4100*0.27 = 1107 $*
*Cost $ = 280 + 70*1 = 350 $
Profit/ha $ = 1107 - 350 = 757 $

Rate = 2 kg/ha

yield = 4200
Revenue $ = 4200*0.27 = 1134 $*
*Cost $ = 280 + 70*2 = 420 $
Profit/ha $ = 1134 - 420 = 714 $

Corn

Rate = 0 kg/ha

yield = 5900
Revenue $ = 5900*0.22 = 1298 $*
*Cost $ = 390 + 70*0 = 390 $
Profit/ha $ = 1298 - 390 = 908 $

Rate = 1 kg/ha

yield = 6700
Revenue $ = 6700*0.22 = 1474 $*
*Cost $ = 390 + 70*1 = 460 $
Profit/ha $ = 1474 - 460 = 1014 $

Rate = 2 kg/ha

yield = 7900
Revenue $ = 7900*0.22 = 1738 $*
*Cost $ = 390 + 70*2 = 530 $
Profit/ha $ = 1738 - 530 = 1208 $

$ x_{i, j} $ = hectares of $ i [S, W, C] $ and $ j [0, 1, 2] $, where $ i $ denotes crop and $ j $ denotes rate of pesticide application.

Maximize $ Z = 2900x_{S,0}+ 3800x_{S,1} + 4400x_{S,2} + 3500x_{W,0} + 4100x_{W,1} + 4200x_{W,2} + 5900x_{C,0} + 6700x_{C,1} + 7900x_{C,2} $

x_{S,0} + x_{S,1} + x_{S,2} + x_{W,0} + x_{W,1} + x_{W,2} + x_{C,0} + x_{C,1} + x_{C,2} $

$ x_{S,1} + 2x_{S,2} 0.8x_{S,0} + 0.8x_{S,1} + 0.8x_{S,2} $
$ 0 0.8x_{S,0} - 0.2x_{S,1} - 1.2x_{S,2} $

$ x_{W,1} + 2x_{W,2} 0.7x_{W,0} + 0.7x_{W,1} + 0.7x_{W,2} $
$ 0 0.7x_{W,0} - 0.3x_{W,1} - 1.3x_{W,2} $

$ x_{C,1} + 2x_{C,2} 0.6x_{C,0} + 0.6x_{C,1} + 0.6x_{C,2} $
$ 0 0.6x_{C,0} - 0.4x_{C,1} - 1.4x_{C,2} $

Constraints: $ x_{S,0} + x_{S,1} + x_{S,2} + x_{W,0} + x_{W,1} + x_{W,2} + x_{C,0} + x_{C,1} + x_{C,2} 130 $
$ 0 0.8x_{S,0} - 0.2x_{S,1} - 1.2x_{S,2} $
$ 0 0.7x_{W,0} - 0.3x_{W,1} - 1.3x_{W,2} $
$ 0 0.6x_{C,0} - 0.4x_{C,1} - 1.4x_{C,2} $
$ 2900x_{S,0} + 3800x_{S,1} + 4400x_{S,2} 250000 $
$ 3500x_{W,0} + 4100x_{W,1} + 4200x_{W,2} 250000 $
$ 5900x_{C,0} + 6700x_{C,1} + 7900x_{C,2} 250000 $
$ x_{i,j} 0 $

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

```
farm_model = Model(HiGHS.Optimizer)
set_silent(farm_model)

@variable(farm_model, x_S0 >= 0)  # soybeans, 0 kg/ha
@variable(farm_model, x_S1 >= 0)  # soybeans, 1 kg/ha
@variable(farm_model, x_S2 >= 0)  # soybeans, 2 kg/ha

@variable(farm_model, x_W0 >= 0)  # wheat, 0 kg/ha
@variable(farm_model, x_W1 >= 0)  # wheat, 1 kg/ha
@variable(farm_model, x_W2 >= 0)  # wheat, 2 kg/ha

@variable(farm_model, x_C0 >= 0)  # corn, 0 kg/ha
@variable(farm_model, x_C1 >= 0)  # corn, 1 kg/ha
@variable(farm_model, x_C2 >= 0)  # corn, 2 kg/ha

@objective(farm_model, Max, 694*x_S0 + 948*x_S1 + 1094*x_S2 + 665*x_W0 + 757*x_W1 + 714*x_W2
```

```
@constraint(farm_model, land_constraint, x_S0 + x_S1 + x_S2 + x_W0 + x_W1 + x_W2 + x_C0 + x_C
@constraint(farm_model, env_soy, 0.8*x_S0 - 0.2*x_S1 - 1.2*x_S2 >= 0)
@constraint(farm_model, env_wheat, 0.7*x_W0 - 0.3*x_W1 - 1.3*x_W2 >= 0)
@constraint(farm_model, env_corn, 0.6*x_C0 - 0.4*x_C1 - 1.4*x_C2 >= 0)
@constraint(farm_model, demand_soy, 2900*x_S0 + 3800*x_S1 + 4400*x_S2 <= 250000)
@constraint(farm_model, demand_wheat, 3500*x_W0 + 4100*x_W1 + 4200*x_W2 <= 250000)
@constraint(farm_model, demand_corn, 5900*x_C0 + 6700*x_C1 + 7900*x_C2 <= 250000)

optimize!(farm_model)

println("Optimal land allocation (hectares):")
println("Soybeans - 0 kg/ha: ", round(value(x_S0), digits=2), " ha")
println("Soybeans - 1 kg/ha: ", round(value(x_S1), digits=2), " ha")
println("Soybeans - 2 kg/ha: ", round(value(x_S2), digits=2), " ha")
println("Total soybeans: ", round(value(x_S0 + x_S1 + x_S2), digits=2), " ha")
println()

println("Wheat - 0 kg/ha: ", round(value(x_W0), digits=2), " ha")
println("Wheat - 1 kg/ha: ", round(value(x_W1), digits=2), " ha")
println("Wheat - 2 kg/ha: ", round(value(x_W2), digits=2), " ha")
println("Total wheat: ", round(value(x_W0 + x_W1 + x_W2), digits=2), " ha")
println()

println("Corn - 0 kg/ha: ", round(value(x_C0), digits=2), " ha")
println("Corn - 1 kg/ha: ", round(value(x_C1), digits=2), " ha")
println("Corn - 2 kg/ha: ", round(value(x_C2), digits=2), " ha")
println("Total corn: ", round(value(x_C0 + x_C1 + x_C2), digits=2), " ha")
println()
```

```
Optimal land allocation (hectares):
Soybeans - 0 kg/ha: 13.81 ha
Soybeans - 1 kg/ha: 55.25 ha
Soybeans - 2 kg/ha: 0.0 ha
Total soybeans: 69.06 ha

Wheat - 0 kg/ha: 6.74 ha
Wheat - 1 kg/ha: 15.73 ha
Wheat - 2 kg/ha: 0.0 ha
Total wheat: 22.48 ha

Corn - 0 kg/ha: 26.92 ha
Corn - 1 kg/ha: 0.0 ha
```

```
Corn - 2 kg/ha: 11.54 ha
Total corn: 38.46 ha
```

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

```
# Solve with 140 hectares to verify
farm_model_140 = Model(HiGHS.Optimizer)
set_silent(farm_model_140)

@variable(farm_model_140, x_S0 >= 0)
@variable(farm_model_140, x_S1 >= 0)
@variable(farm_model_140, x_S2 >= 0)
@variable(farm_model_140, x_W0 >= 0)
@variable(farm_model_140, x_W1 >= 0)
@variable(farm_model_140, x_W2 >= 0)
@variable(farm_model_140, x_C0 >= 0)
@variable(farm_model_140, x_C1 >= 0)
@variable(farm_model_140, x_C2 >= 0)

@objective(farm_model_140, Max, 694*x_S0 + 948*x_S1 + 1094*x_S2 + 665*x_W0 + 757*x_W1 + 714*

@constraint(farm_model_140, land_constraint_140, x_S0 + x_S1 + x_S2 + x_W0 + x_W1 + x_W2 + x
@constraint(farm_model_140, env_soy_140, 0.8*x_S0 - 0.2*x_S1 - 1.2*x_S2 >= 0)
@constraint(farm_model_140, env_wheat_140, 0.7*x_W0 - 0.3*x_W1 - 1.3*x_W2 >= 0)
@constraint(farm_model_140, env_corn_140, 0.6*x_C0 - 0.4*x_C1 - 1.4*x_C2 >= 0)
@constraint(farm_model_140, demand_soy_140, 2900*x_S0 + 3800*x_S1 + 4400*x_S2 <= 250000)
@constraint(farm_model_140, demand_wheat_140, 3500*x_W0 + 4100*x_W1 + 4200*x_W2 <= 250000)
@constraint(farm_model_140, demand_corn_140, 5900*x_C0 + 6700*x_C1 + 7900*x_C2 <= 250000)

optimize!(farm_model_140)

current_profit = objective_value(farm_model)
new_profit = objective_value(farm_model_140)
profit_increase = new_profit - current_profit

println("\nVerification with 140 hectares:")
println("New optimal profit: \$", round(new_profit, digits=2))
println("Actual profit increase: \$", round(profit_increase, digits=2))
println("Profit increase per hectare: \$", round(profit_increase/10, digits=2))
```

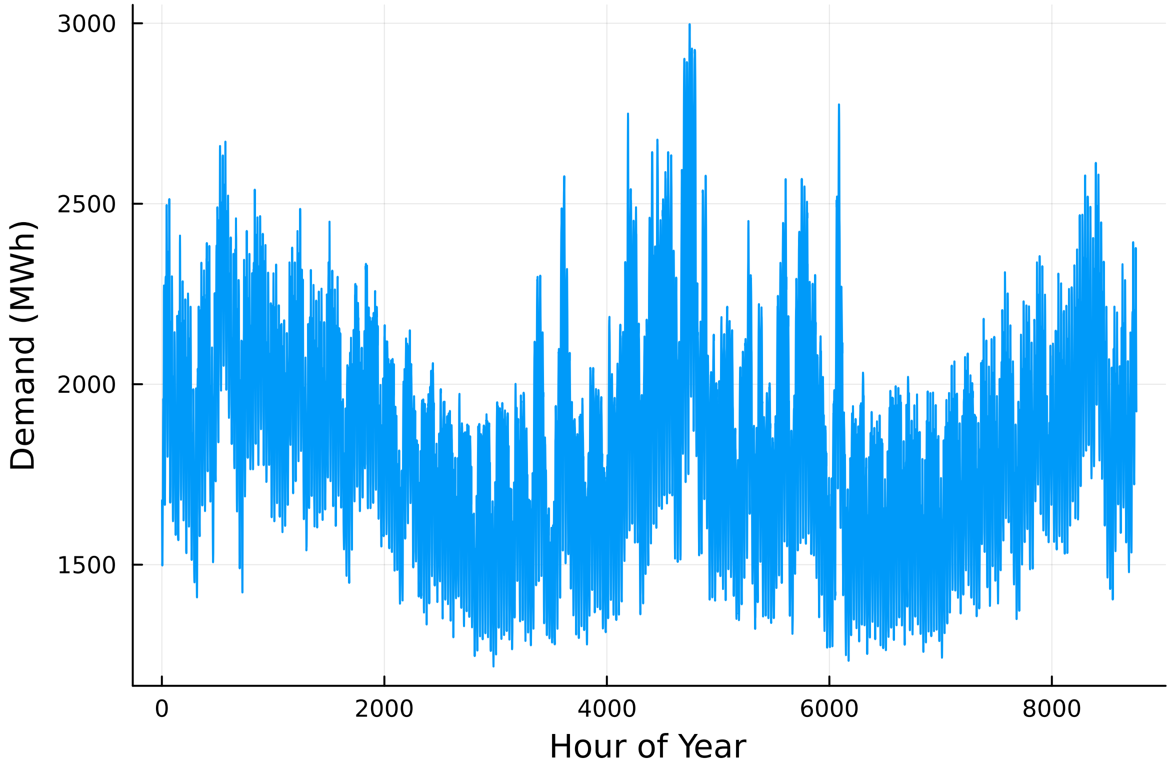
```
Verification with 140 hectares:
New optimal profit: $124035.17
Actual profit increase: $7294.0
Profit increase per hectare: $729.4
```

Problem 3 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, :"Time Stamp" => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```



Next, we load the generator data, shown in Table 3. This data includes fixed costs (\$/MW installed), variable costs (\$/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```
# load capacify factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

display(p1)
display(p2)
```

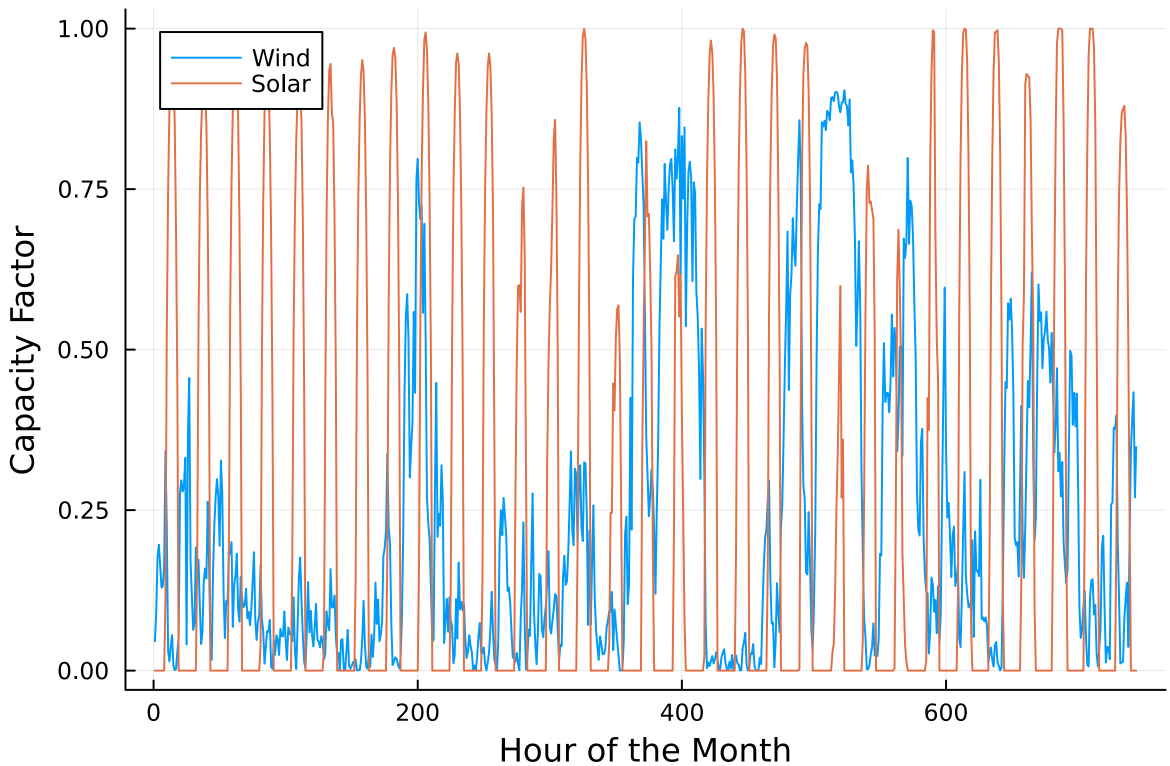
You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO2 limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO2/yr.

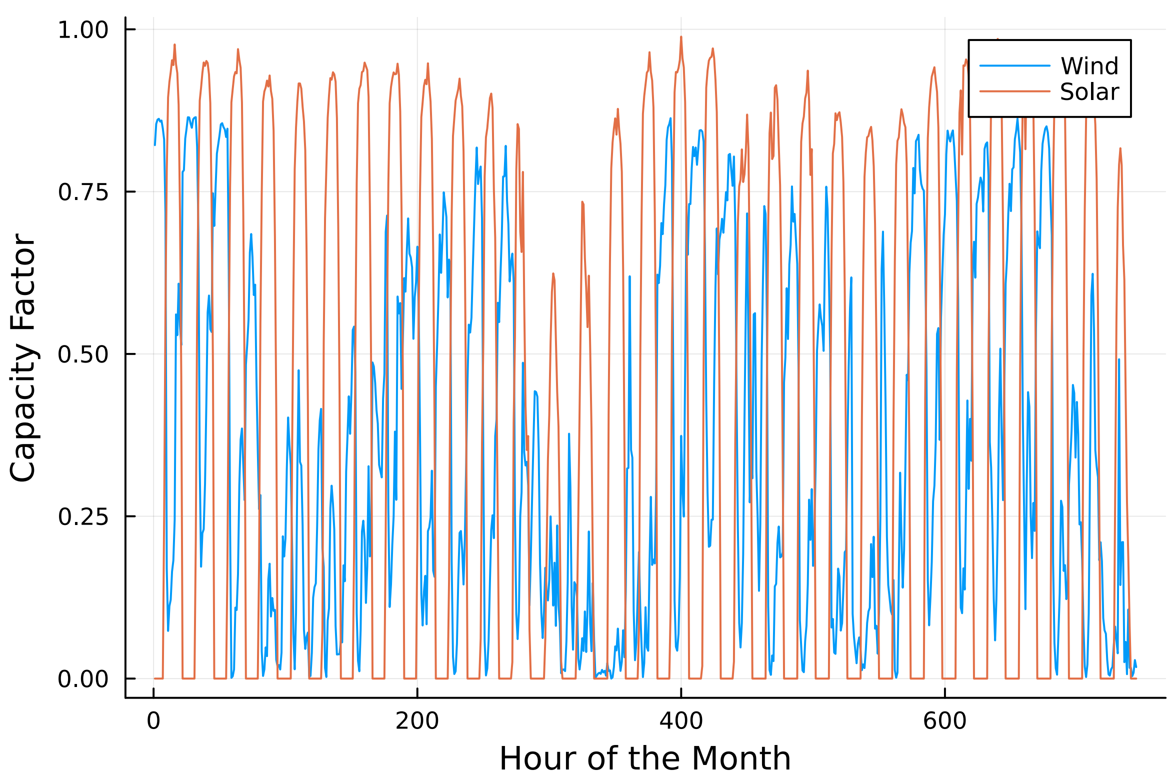
While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of $10,000/MWh.

Problem 3.1

Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

G = geothermal
C = coal
CCGT = natural gas combined cyclone gas turbine
CT = natural gas combustion turbine
W = wind
S = solar


Wind
Solar
Capacity Factor
1.00
0.75
0.50
0.25
0.00
0
200
400
600
Hour of the Month


Wind
Solar
Capacity Factor
1.00
0.75
0.50
0.25
0.00
0
200
400
600
Hour of the Month

$ x_g $ = installed capacity (MWh) from generator type g $ $ [G, C, CCGT, CT, W, S]
$ y_{g,t} $ = production (MWh) from each generatore type g in hour t
$ NSE_t $ = non served energy (MWh) in hour t
$ c_g = CO_2 $ emissions for generator (tCO_2/MWh) type g
$ d_t $ = energy demand in hour t

$ FX_g $ = fixed costs for generator type g
$ VC_g $ = variable costs for generator type g
$ NSEC = 10000 $/MWh $ non-served energy (MWh) cost

Minimize investment cost + operating cost $ = {g} FX_g * x_g + {g} {t} VC_g * y{g, t} + _{t} NSEC * NSE_t $

Constraints:
$ y_{G,t} + y_{C,t} + y_{CCGT,t} + y_{CT,t} + y_{W,t} + y_{S,t} + NSE_t d_t $
$ y_{g,t} x_g $
$ {t} y{G,t}c_{G} + y_{C,t}c_{C} + y_{CCGT,t}c_{CCGT} + y_{CT,t}c_{CT} + y_{W,t}c_{W} + y_{S,t}c_{S} 15,000,000 $
$ y_{g,t}, x_g 0 $

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?

```
generators = ["Geothermal", "Coal", "NG CCGT", "NG CT", "Wind", "Solar"]

G = 1:nrow(gens[1:end, :])
T = 1:nrow(demand)
NSECost = 10000
CO2_max = 1_500_000

CF = Dict(
    1 => fill(0.85, length(T)),
    2 => fill(1.0, length(T)),
    3 => fill(1.0, length(T)),
    4 => fill(1.0, length(T)),
    5 => cap_factor.Wind,
    6 => cap_factor.Solar
)

gencap = Model(HiGHS.Optimizer)
set_silent(gencap)
```

```
@variable(gencap, x[g in G] >= 0)
@variable(gencap, y[g in G, t in T] >= 0)
@variable(gencap, NSE[t in T] >= 0)

fc = Dict(g => gens.FixedCost[g] for g in G)
vc = Dict(g => gens.VarCost[g] for g in G)
e = Dict(g => gens.Emissions[g] for g in G)

@objective(gencap, Min, sum(fc[g] * x[g] for g in G) + sum(vc[g] * y[g,t] for g in G, t in T

#Must meet demand
@constraint(gencap, load[t in T], sum(y[:, t]) + NSE[t] == demand.Demand[t])
#Output must not exceed maximum efficiency output for each plant
@constraint(gencap, max_output[g in G, t in T], y[g, t] <= x[g]*CF[g][t])
#Must meet CO2 emissions limit
@constraint(gencap, co2_limit, sum(e[g] * sum(y[g, t] for t in T) for g in G) <= CO2_max)

optimize!(gencap)
```

```
for g in G
    capacity = value(x[g])
    println("$(generators[g]): ", round(capacity, digits=2), " MW")
end

println("-"^30)

total_cost = objective_value(gencap)
println("Total cost: \$", Int(round(total_cost)))

# Break down the cost components
fixed_cost = sum(gens[g, :FixedCost] * value(x[g]) for g in G)
variable_cost = sum(gens[g, :VarCost] * sum(value(y[g, t]) for t in T) for g in G)
nse_cost = NSECost * sum(value(NSE[t]) for t in T)

println("Cost breakdown:")
println("- Fixed costs: \$", Int(round(fixed_cost)))
println("- Variable costs: \$", Int(round(variable_cost)))
println("- Non-served energy costs: \$", Int(round(nse_cost)))

println("-"^30)

total_nse = sum(value(NSE[t]) for t in T)
```

```julia
println("Total non-served energy: ", Int(round(total_nse)), " MWh")
```

```
Geothermal: 285.57 MW
Coal: 0.0 MW
NG CCGT: 1509.17 MW
NG CT: 703.06 MW
Wind: 2548.89 MW
Solar: 2103.75 MW
------------------------------
Total cost: $785352959
Cost breakdown:
- Fixed costs: $677168118
- Variable costs: $104861933
- Non-served energy costs: $3322907
------------------------------
Total non-served energy: 332 MWh
```

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

```julia
println("Energy generation by plant type:")

total_gen = sum(value(y[g, t]) for g in G, t in T)

for g in G
    total_gen_frac = (sum(value(y[g, t]) for t in T))/total_gen
    println("$(generators[g]): ", round(total_gen_frac*100, digits=2), "%")
end
```

```
Energy generation by plant type:
Geothermal: 6.7%
Coal: 0.0%
NG CCGT: 20.3%
NG CT: 0.79%
Wind: 39.42%
Solar: 32.79%
```

The vast majority of energy provided comes from renewable energies (Geothermal, Wind, Solar sum to 78.91%). This shows the impact of the CO2 constraint, limiting emissions to no more than 1.5 megatons of CO2. There is still some use of non-renewable energies, with 20.3% coming from NG CCGT and 0.79% coming from NG CT, which fills in the gaps for when renewable energies (particularly wind and solar) are not able to meet demand due to varying capacity factors.

This makes sense, and falls in line with the 3.2 results. The capacity for Wind and Solar are highest and second highest, showing their use is most important. Goethermal, having no variability in output, means that while it can be useful, it's capacity needs to remain low to avoid excess production costs when solar and wind are at high capacity factor. However, because natural gas generators can change production output at will, they have a higher capacity than geothermal but lower than wind and solar to be able to have high generation rates during shorter periods of time to fill the demand gap.

Problem 3.5

What is the shadow price of CO2? What is the value to the utility be of allowing it to emit an additional 1000 tCO2/yr?

> **Significant Digits**
>
> Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.
>
> **Getting Variable Output Values**
>
> `value.(x)` will report the values of a `JuMP` variable `x`, but it will return a special container which holds other information about `x` that is useful for `JuMP`. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.
>
> **Suppressing farm_model Command Output**
>
> The output of specifying farm_model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing output by adding a semi-colon after the last command, or you might find that your notebook crashes.

```
co2_shadow_price = abs(dual(co2_limit))

value_additional_1000t = co2_shadow_price * 1_000

println("Shadow price of CO  constraint: \$", round(co2_shadow_price; digits=4), " per kg CO
println("Value of allowing additional 1000 tCO /yr: \$", Int(round(value_additional_1000t)))

# Additional context about the current CO2 emissions
total_co2 = sum(e[g] * sum(value(y[g, t]) for t in T) for g in G)
co2_used = total_co2 / CO2_max * 100

println("Current CO  emissions: ", Int(round(total_co2)), " kg CO ")
println("CO  limit utilization: ", round(co2_used), "%")
```

```
Shadow price of CO  constraint: $182.7719 per kg CO
Value of allowing additional 1000 tCO /yr: $182772
Current CO  emissions: 1500000 kg CO
CO  limit utilization: 100.0%
```

References

List any external references consulted, including classmates.
Got some help from Tómas and Pranati in figuring out some issues with my linear program for problem 3.2